



Management of Electronic Waste in Sub-Saharan Africa: Economic and Environmental Challenges

REVIEW PAPER

Groupe BG

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Introduction

In 2021, the global generation of electronic waste reached approximately 57.4 million tons¹, a number expected to escalate in the coming years. The surge in e-waste is predominantly attributed to planned obsolescence, a strategy intentionally shortening the lifespan of Electrical and Electronic Equipment (EEE). For instance, computers, which once had a lifespan of 5-10 years, now endure a mere 3-4 years (Maphosa, V. and Maphosa, M. 2020, p. 3). Moreover, there is a prevailing trend that favors replacement over repair, driven by deliberate design choices like non-replaceable smartphone batteries.

Asante, et al. (2019, p. 115) state that merely around 15% of this massive waste is believed to undergo comprehensive recycling. Despite Africa contributing the lowest regional amount of e-waste globally, approximately 2.9 million metric tons², it contends with some of the largest e-waste dumpsites on a global scale. This paradox is compounded by the fact that roughly 80% of e-waste from developed nations is illicitly exported to developing countries, according to Williams and Adetuyi (2022, p.1) , with 60% of this waste finding its way to African nations—a phenomenon commonly termed as leakage.

This review paper will first delve deeper into the economic repercussions in three heavily impacted sub-Saharan countries: Ghana, South Africa, and Nigeria. As seen in figure 1 below, the southern African region dominates in terms of electronic waste recycling and collection, followed by West African countries in third position.

Subsequently, this paper will elucidate the environmental and health framework concerning e-waste management and conclude by highlighting the absence of specific protective policies in these three countries.

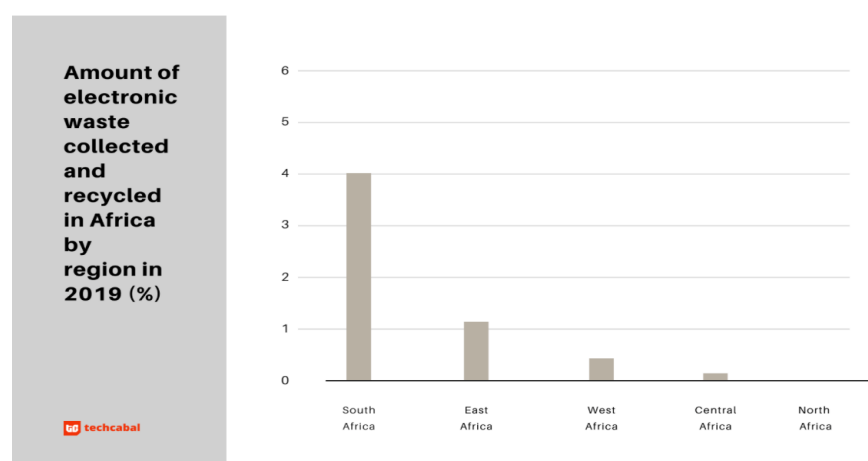


Figure 1. Amount of electronic waste collected and recycled in Africa by region in 2019 (source techcabal)

Source: The Global E-waste Monitor 2020

¹ Williams, N. and Adetuyi, A. (2022). 'E-Waste Management in Africa - Overview and Policy Developments'. [online] *Brooks and Knights Legal Consultants*. Available at:

<https://brooksandknights.com/2022/03/24/e-waste-management-in-africa-overview-and-policy-developments/#:~:text=About%2080%25%20of%20the%20e>

² ibid

Economic Impact of E-Waste in Sub-Saharan Africa

In the economies of Ghana, Nigeria, and South Africa, the informal e-waste sector serves as a significant source of employment. Agbogbloshie in Ghana and Alaba in Nigeria stand as primary sites where informal e-waste recycling practices prevail, utilizing basic methods to salvage valuable materials. Conversely, South Africa boasts a limited number of formal recycling facilities, while informal practices persist. This informal sector serves as a dual-edged sword: it offers livelihoods but exposes workers to health hazards and contributes to environmental degradation.

The e-waste recycling industry, valued at 60 billion dollars in 2023 (Alves, B., 2023), has a significant economic potential, particularly in the recovery of precious metals like gold, silver and copper. Notably, the Agbogbloshie site in Ghana plays a role in refurbishing about 25% of donated computers to schools, providing workers with incomes to meet basic needs. Regarding Nigeria, over 30,000 e-waste recyclers contribute to the GDP (Williams and Adetuyi, 2022, p.6). Agbogbloshie directly employed over 15,000 workers, while informal workers in Ghana recovered valuable metals like gold, copper, and silver from e-waste, contributing to their livelihoods³. Despite this potential, e-waste workers earn meager incomes due to the lack of regulations in this informal sector. Collectors, crucial to the e-waste value chain, earn approximately USD4 per day, a competitive income compared to USD1.40 which is earned by other informal workers in Ghana.⁴

In these nations, the informal e-waste economy plays a pivotal role in generating both direct and indirect employment opportunities. In Ghana, e-waste-related activities generated between 105 and 268 million US dollars in 2014, providing jobs for at least 200,000 individuals across the country.⁵ Agbogbloshie, recognized as sub-Saharan Africa's largest e-waste site, creates 4,000 to 6,000 direct jobs⁶, fostering entrepreneurial vigor and economic prospects through community-based collection, recovery, and recycling. Furthermore, South Africa displays a blend of formal and informal e-waste management activities, offering employment avenues. The informal sector contributes to 25% of e-waste recycling in South Africa, employing roughly 10,000 individuals, including 2,000 regular workers⁷. Despite its contribution, this sector faces challenges such as health risks and uncertain working conditions. However, South Africa's transition towards a circular economy and formal recycling also generates employment opportunities in safer sectors.

The economic costs associated with e-waste management in Ghana, Nigeria, and South Africa highlight varying aspects within these nations. These costs encompass public health expenses, environmental deterioration, and direct or indirect economic losses. In Ghana, the

³Maphosa, V. and Maphosa, M. (2020). 'E-waste management in Sub-Saharan Africa: A systematic literature review'. *Cogent Business & Management*, 7(1) [online]. Available at :<https://doi.org/10.1080/23311975.2020.1814503> (accessed: 22.11.2023).

⁴ Ibid.

⁵ Maes, T. and Preston-Whyte, F. (2022). 'E-waste it wisely: lessons from Africa'. *SN Applied Sciences*, 4(3) [online]. <https://doi.org/10.1007/s42452-022-04962-9> (accessed: 22.11.2023).

⁶ Ibid.

⁷ Ibid.

economic impact stems from Agbogbloshie's environmental degradation, leading to substantial healthcare costs linked to e-waste exposure. Similarly, Nigeria incurs significant economic costs due to health and environmental repercussions, including income losses and land devaluation. In South Africa, the economic impact involves higher expenditures for waste management and site rehabilitation, alongside substantial direct health and environmental costs. Although direct numerical comparisons between these countries are challenging due to data constraints, all three experience considerable economic implications from inadequate e-waste management, necessitating measures to mitigate these costs and promote more sustainable e-waste practices.

Resource recovery from e-waste in countries like Ghana, Nigeria, and South Africa predominantly relies on informal practices. Workers in precarious conditions manually extract precious metals like copper and aluminum from electronic devices. The absence of proper recycling infrastructure limits efficient resource recovery, causing missed economic opportunities. To maximize the economic potential of e-waste, effective policies, investments in modern recycling infrastructure, and advanced technologies are necessary. This would also help improve the health and environmental risks associated with e-waste recycling practices.

Environmental and Health Framework of E-Waste Management

Developing African nations such as Nigeria, Ghana and South Africa are facing significant environmental and health challenges because of the electronic waste being dumped there. These countries receive damaged electronic devices such as phones, laptops, batteries, televisions, home appliances and more from developed countries that don't need them anymore. The countries that contribute the most to the e-waste found in Africa are China (24%), the US (20%), Spain (12%) and the UK (9%)⁸. This is because of the large demand for new electronic devices in developed countries.

The mismanagement of the e-waste dumped in African regions has a negative impact on our environment. Countries like Nigeria, Ghana and South Africa have seen negative changes in their big cities such as Lagos, Accra and Cape town. This is because the processing of e-waste is mostly done in open backyards where parts are manually dismantled, smashed and burned. This process is done because e-waste contains materials of important value and by recycling them profits can be made. Therefore, workers try to retrieve any valuable rare-earth metals which are resold and reused and unwanted parts are thrown into large open dumpsites. The improper handling of these gadgets by untrained workers create cancerous pollutants like lead, cadmium, mercury, lithium, nickel, silver, palladium and zinc⁹. Additionally, e-waste

⁸ Lebbie, T.S., Moyebi, O.D., Asante, K.A., Fobil, J., Brune-Drisse, M.N., Suk, W.A., Sly, P.D., Gorman, J. and Carpenter, D.O. (2021). 'E-Waste in Africa: A Serious Threat to the Health of Children'. *International Journal of Environmental Research and Public Health*, 18(16) [online]. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8392572/>

⁹ Rautela, R., Arya, S., Vishwakarma, S., Lee, J., Kim, K.-H. and Kumar, S. (2021). 'E-waste management and its effects on the environment and human health'. *Science of The Total Environment*, [online] available at: <https://doi.org/10.1016/j.scitotenv.2021.145623>.

includes plastics, organic chemicals and rare earth metals, many of which have unknown health effects.

As a result, the air around these waste sites is significantly toxic but workers and residents have no other choice but to inhale it. These toxins also impact the environment by contaminating dust, soils and water. The surrounding vegetation is negatively impacted as well as many aquatic species such as fish. This makes the food and water residents have access to very polluted and unsafe. Furthermore, the process of burning cooling devices such as refrigerators impacts the ozone layer and consequently contributes to climate change.

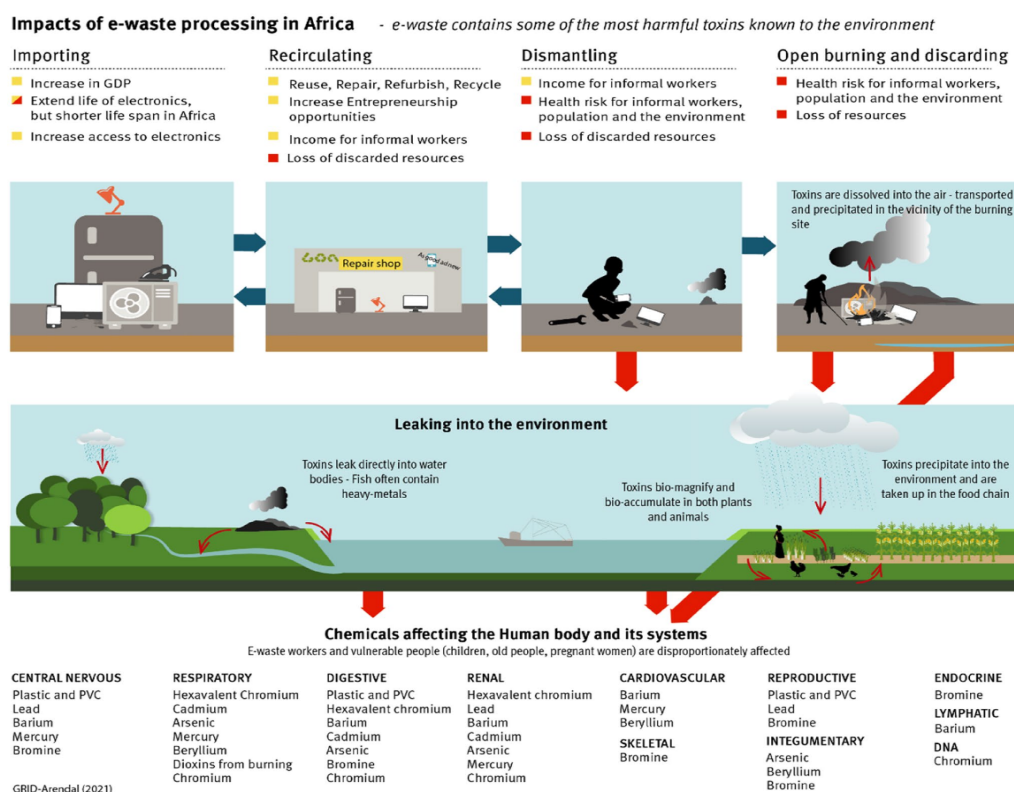


Figure 2. Impact of e-waste processing in Africa (source: science direct)

In addition to the many environmental risks related to e-waste there are also numerous adverse health outcomes. The toxins from the e-waste that workers are constantly in contact accumulate in their bodies and often lead to health issues. They suffer from eye problems because of the fumes and chemicals released when burning waste. Furthermore, some workers develop carcinogenic, mutagenic or genotoxic conditions¹⁰. Other health issues encountered by workers include respiratory problems, skin diseases, physical injuries, malaria and depression.

Women and babies that do not work on the dump sites are also indirectly susceptible to the dangerous toxins emanating from the e-waste sites. While there is a significant exposure risk

¹⁰ Maes, T. and Preston-Whyte, F. (2022). 'E-waste it wisely: lessons from Africa'. *SN Applied Sciences*, 4(3) [online]. <https://doi.org/10.1007/s42452-022-04962-9> (accessed: 22.11.2023).

for all women, pregnant women face a heightened vulnerability to the pollutants from e-waste recycling activities. Expectant mothers with significant exposure to e-waste toxins have high chances of giving birth to a baby with impaired neurological and behavioral development.

Furthermore, numerous children as young as 5 years old are involved in unregulated e-waste activities in Africa¹¹, where they pick, burn and dismantle electrical items. As these young children are still developing they are at a heightened risk when exposed to hazardous chemicals. Their small bodies face challenges in breaking down and eliminating toxins compared to adults. Additionally, due to their growing bodies they are much more sensitive to pollutants. Therefore, the negative impacts to their system may result in enduring health consequences¹².

Challenges of E-Waste Management in Sub-Saharan Africa

The African continent has the least coverage in terms of e-waste legislation. In fact, no country in Sub-Saharan Africa has specific laws; they mostly rely on legislation or regional strategies that address the management of hazardous waste. Among the three countries forming the basis of our research, South Africa has implemented the most extensive legislation and has taken significant steps in managing e-waste.

There are two primary conventions that address the regulation of e-waste. The first is the Basel Convention, which came into force in 1992 in response to increasing concerns about the inappropriate management of e-waste as hazardous waste across national borders in Africa¹³. Since its inception, the international convention has prohibited the export of hazardous waste from one country to another unless it can be reused or repaired. However, this condition led to widespread exceptions, and unfortunately, practical enforcement has been lacking. The deficiency in enforcement prompted the addition of several Annexes in 2019 to maintain the relevance and effectiveness of the convention. These amendments aimed to improve the control of transboundary movement of hazardous plastic waste across Africa. The Basel Convention has been ratified by 191 countries including Ghana, Nigeria and South Africa.¹⁴

The second agreement is the Bamako Convention, established in 1998, which is a regional treaty consisting among African countries prohibiting the import of any hazardous waste in

¹¹ Lebbie, T.S., Moyebi, O.D., Asante, K.A., Fobil, J., Brune-Drise, M.N., Suk, W.A., Sly, P.D., Gorman, J. and Carpenter, D.O. (2021). 'E-Waste in Africa: A Serious Threat to the Health of Children'. *International Journal of Environmental Research and Public Health*, 18(16) [online]. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8392572/>

¹² World Health Organization (2023). *Electronic waste (e-waste)*. [online]. Available at: [https://www.who.int/news-room/fact-sheets/detail/electronic-waste-\(e-waste\)](https://www.who.int/news-room/fact-sheets/detail/electronic-waste-(e-waste)).

¹³ Maes, T. and Preston-Whyte, F. (2022). 'E-waste it wisely: lessons from Africa'. *SN Applied Sciences*, 4(3) [online]. <https://doi.org/10.1007/s42452-022-04962-9>

¹⁴ UN environment program Basel convention. (n.d.). *Parties to the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal*. [online] Available at: <https://www.basel.int/Countries/StatusofRatifications/PartiesSignatories/tabid/4499/Default.aspx>

Africa¹⁵. While 29 of 54 African countries have ratified the Convention¹⁶, notable absences include Nigeria, Ghana, and South Africa conversely to the Basel Convention. This highlights the challenge of enforcing laws that ban the importation of e-waste.

Despite this, African countries are still making concerted efforts in addressing e-waste. In 2020, thirteen countries adopted policies and legislations on e-waste, marking a significant increase from three in 2018. As Ghana and Nigeria, between them, host most of the electronic waste, they had to set up the highest regional coverage of electronic waste legislation. An illustrative example of policy efforts reported by Maphosa, V. and Maphosa, M. (2020, p. 11) is Ghana's Act 917, implemented in 2016, to support e-waste management. However, due to insufficient environmental management, the consequences have been the continued arrival of containers filled with electronic waste at various dumpsites every month.

In contrast, significant disparities characterize the policies and management strategies between the three nations. Ghana, for example, funds its management plan through a recycling eco fee imposed on imported EEE, a model initiated since October 2018 (Asante et al, 2019, p.112). In contrast, South Africa adopts a Responsible Producer model (PRO) leaning towards formal mechanisms, aligning with the practices of developed countries, albeit at a higher cost and requiring more substantial investment. However, despite the Ghanaian government's intention to upgrade the Agbogbloshie site in 2018 by establishing a world-class e-waste recycling plant, e-waste management in Ghana and Nigeria remains predominantly informal, particularly at Agbogbloshie.

As stated by Maes and Preston-Whyte (2022, p. 4) Nigeria is a leading importer of used EEE via its major ports in Lagos and holds a strategic position for developed countries to send their waste. Despite formal mechanisms, low coordination between agencies like port and environmental protection officials exacerbates the challenge. Despite efforts to implement laws regulating e-waste, compliance is not always consistent. For instance, compliance with e-waste laws remains inconsistent. A study conducted by Sindiku et al. in Nigeria reveals that nearly half of the stored cathode ray tubes (CRTs) contain hazardous substances such as PBDE, exceeding the limits set by the Restriction of Hazardous Substances directive¹⁷.

Furthermore, it is important to highlight that African countries have modeled their policies on those implemented in developed regions like Europe. Within the European Union, there are few legislative instruments, and companies are striving to enforce new environmental protection models due to environmental pressure and climate change. Regular policies such as the *polluter pays principle* and the *waste hierarchy* are gaining prominence.

¹⁵ Maes, T. and Preston-Whyte, F. (2022). 'E-waste it wisely: lessons from Africa'. *SN Applied Sciences*, 4(3) [online]. <https://doi.org/10.1007/s42452-022-04962-9>

¹⁶ UN Environment. (2020). *Les parties à la convention de Bamako réaffirment leur engagement à protéger l'Afrique contre le déversement de déchets dangereux*. [online] Available at:

<https://www.unep.org/fr/actualites-et-recits/communiqu%C3%A9-de-presse/les-parties-la-convention-de-bamako-reaffirment-leur>

¹⁷ Asante, K.A., Amoyaw-Osei, Y. and Agusa, T. (2019). 'E-waste recycling in Africa: risks and opportunities'. *Current Opinion in Green and Sustainable Chemistry*, 18 [online], p.114. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S2452223618301445>

Most European policies are built on the principle of Extended Producer Responsibility (EPR), a waste management model that places the responsibility on producers of Electrical and Electronic Equipment to collect and treat the wastes generated by their products (Williams and Adetuyi, 2022, p.5). EPR encourages producers to design and manufacture EEE while considering lifecycle issues such as repair, upgrading and recycling.

Implementing the EPR framework in the African context can be challenging because importation is more common in countries like Ghana or Nigeria rather than manufacturing. This is mainly due to inadequate or excessively costly infrastructure, as well as a shortage of technology or financing.

Therefore, it is crucial for African regulatory frameworks to continue their mission to ensure that the e-waste management system can improve and find alternative models for e-waste regulations. This should be possible with the proper support from their respective governments and non-governmental organizations.

Conclusion and Discussion

The effective management of electronic waste in Sub-Saharan Africa presents major environmental and economic challenges. This review highlights the importance of rethinking e-waste management from a sustainable perspective in order to ensure a cleaner and more prosperous future for African regions. It underscores the importance of specific policies, adequate infrastructure, and innovative recycling practices to mitigate the risks to the environment and human health while harnessing the economic potential of e-waste. Through the implementation of strict regulations on e-waste movements, it would be possible to reduce massive import flows, enabling a more formalized recycling system. However, as mentioned in the review, it remains challenging to find investment funds willing to finance these projects, especially in countries like Ghana and Nigeria.

Nevertheless, a significant challenge in this situation is to implement regulations that do not compromise the economic stability of the importing countries, ensuring, for instance, that the regulations do not lead to the elimination of numerous jobs. Such a scenario could escalate the unemployment rates in all three countries, potentially leading to political issues. Therefore, there is a need to explore the possibilities of transforming e-waste into a lucrative venture for these countries. The key question is how these nations can strategically leverage e-waste for economic benefit, ensuring a positive impact on both the environment and the workforce.

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